

LEVEL 26 INTEGRATED CONDITIONAL METHODOLOGY — TATE, MAZUR, RIBET, AND KOLYVAGIN IN LEAN 4.12

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ABSTRACT. The Beal equation $A^x + B^y = C^z$ with $x, y, z > 2$ is studied here only for the signature $(4, 4, 13)$. The associated Frey curve $E : y^2 = x(x - A^4)(x + B^4)$ has displayed conductor $928 = 2^5 \cdot 29$, Néron types I_0^* at 2 and I_8 at 29, invariants c_4 and Δ , and valuations $v_2(\Delta) = 6$, $v_{29}(\Delta) = 8$. This draft records the Lean 4.12 structural verification of four honest formal surfaces and their conditional integration. Mazur: $|\mathrm{SL}_2(\mathbb{F}_{13})| = 2184$, $48 < 2184$, $13 = 2^2 + 3^2$, $288/48 = 6$, cusp Finset card 2, genus 0 (so $X_0(13)(\mathbb{Q})$ is infinite). Ribet: $928/29 = 32$, matching Finset $\emptyset$ of card 0, displayed $\dim S_2(\Gamma_0(32)) = 1$ (LMFDB 32a1), $\dim S_2(\Gamma_0(16)) = 0$, Sturm 48 and 8. Kolyvagin: rank-0 display on Cremona 26a1/26b1 with $|\mathrm{Sel}_2| = 1$, $3 \cdot 7 = 21$, $L/\Omega = 1/3$ and $1/7$, and $\mathrm{Nonempty}(\mathrm{Fintype} \alpha) \rightarrow \mathrm{Finite} \alpha$. The continued Tate boundary proves the discriminant and valuation numerics, records I_{26k} for $k = v_{29}(B+3)$ only as the output of actual Tate-algorithm evidence, and orders Néron 32/928, cyclotomic $q^{1/13}$ inertia, and mod-13 unramifiedness strictly forward. The continued Mazur boundary likewise orders point-level $X_0(13)$ data, the exact Fricke function, attached Frey $j = c_4^3/\Delta$, Borel reducibility, and global exclusion strictly forward. The continued Ribet boundary orders modularity, the attached residual representation and irreducibility, Tate–Néron conductor data, level lowering, 32a1 identification, the $a_{29} = -10$ mismatch, and complete bad-reduction elimination strictly forward. The continued Kolyvagin boundary orders curve-attached L -functions and periods, normalized-value identifications, Kato–Kolyvagin, finite generation, constructed Selmer exact sequences, cardinality identifications, and actual **Fintype** witnesses strictly forward. The final Conditional-32 boundary orders the concrete Mazur, Tate, and Ribet conclusions, three explicit typed conversions to the older abstract interface, and **HasNewformAtLevel32** strictly forward. The final **RequiresTateMazurRibetFor32.ForwardDoc** table preserves the Tate, Néron/inertia, Mazur, Ribet, Kolyvagin, and Conditional-32 interfaces in that exact order while retaining the three typed conversions. The final **BealFinal.Main** interface adds the common-prime conclusion only as explicit proof data and keeps the full $B \leq 10^6$ search separate from the proved $B < 1000$ slice. The post-final v25 retrofit orders the Matveev bound, level-26 arithmetic, the proved small search, and the explicit full-search boundary without using numerics backwards. The integrated bridge derives proposition-valued level-32 evidence from exactly three explicit Mazur, Tate, and Ribet assumptions; it does not construct an unconditional Mathlib newform. The new final module adds no **native_decide**; its numerical theorem inherits the existing finite-search certificate. Software version DOI 10.5281/zenodo.22922473; concept DOI 10.5281/zenodo.22379293.

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1. INTRODUCTION

The Beal conjecture asserts that if $A^x + B^y = C^z$ with integers $A, B, C \neq 0$ and exponents $x, y, z > 2$, then A, B, C share a prime factor [10]. This draft does not prove the conjecture. It records a *structural verification* of four honest Lean 4.12 displays used in the level-26 attack on the signature (4, 4, 13),

$$A^4 + B^4 = C^{13}.$$

The associated Frey curve is the displayed Weierstrass model

$$y^2 = x(x - A^4)(x + B^4) \quad \text{i.e.} \quad [0, B^4 - A^4, 0, -A^4B^4, 0]. \quad (1)$$

The short model $[0, -(A^4 + B^4), 0, 0, 0]$ is a different curve and is not used as E .

The strategy is classical [7, 13, 12, 11]: compute the conductor of (1) by Tate’s algorithm [14], exclude a rational 13-isogeny by Mazur’s theorem on $X_0(13)$, lower the residual level $928 \rightarrow 32$ in the style of Ribet, and compare the resulting newform space with the Mordell–Weil data of $J_0(26)$ via the Cremona labels 26a1 and 26b1 [1, 9]. Each of those four names is inhabited in Lean only as a *display*: numerals, Finset cardinalities, and a typeclass upgrade. Parent files keep the corresponding algebraic statements as uninhabited proposition boundaries. The integrated release does not erase those boundaries. Its bridge combines proved Frey numerics with three named assumptions: absence of a rational 13-isogeny, identification of the conductor at 29 with the Néron conductor, and Ribet lowering from 928 to 32. The conclusion is a proposition-valued evidence record, not a modular-form object.

The formal pin is Lean v4.12.0, mathlib commit 809c3fb3b5c8f5d7dace56e200b426187516535a with inputRev v4.12.0 (not @ main), and the beal-conjecture Lake manifest prefix 83739542. The command lake update was not run. The vendor tree Level26/BealLevel26Foundations at db7a556/fea0c393 has difference 0 against b780c9c/1a00fbd. The integrated software version DOI 10.5281/zenodo.22922473 records the post-final retrofit tag v25.0.1-forward-43735b3; the concept DOI 10.5281/zenodo.22379293 stays, related as isVersionOf [3, 2]. The preceding software version is 10.5281/zenodo.22912430.

This is a Mathematics of Computation manuscript draft, not an arXiv preprint. There is no arXiv identifier and no arXiv badge.

The preceding four display files live in Level26/HonestB0Search/ on beal-conjecture main at ea261b9 (badge parent ff546da, merge af520d3) and on foundations phase-darmon-merel-4413 at 2b60ccd. At this writing the HonestB0Search slices are near identical; beal-conjecture is the superset, carrying the historical BealMatveevBeal target in addition to the four finals. The continued companion release is commit 995b9c4, annotated tag v30.1.7-beal-final-43735b3, with exact merge base 43735b3 [6]. The v25 forward addendum is commit 385ff85, annotated tag v25.0.1-forward-43735b3, on that final release [5].

2. TATE — NÉRON CONDUCTOR 928

The file Tate_Frey_Conductor_29_Neron_final.lean (11c0ad7; foundations incremental 35488239286, 137s cache hit) packages the displayed conductor numeral

$$N_E^{\text{disp}} = (2 : \mathbb{N})^5 \cdot 29 = 928.$$

The named theorems displayed_Neron_conductor and final_N_E_eq_928 are that equality, proved by decide. The split displayed_928_div_29 is $928/29 = 32$.

Display 2.1 (Displayed Néron conductor). In Lean 4.12, with `open Nat Finset Classical`,

$$(2 : \mathbb{N})^5 \cdot 29 = 928, \quad 928/29 = 32, \quad 32 \cdot 29 = 928, \quad -29 \mid 32.$$

These are numerals. Mathlib 4.12 has no `KodairaType` and no `NeronModel`. The name $N_E = 928$ is not the conductor of a Néron model.

The displayed Weierstrass invariants of (1) are

$$c_4 = 16(A^8 + B^8 + A^4B^4), \quad \Delta = 16A^8B^8(A^4 + B^4)^2.$$

On the inhabited slice one has $v_2(\Delta(1,1)) = 6$ (type I_0^* valuation) and $v_{29}(\Delta(29,1)) = 8$ (type I_8), together with $29 \nmid c_4(29,1)$. The residual-level Finset $\{32, 928\}$ has card 2; that card is a Finset numeral, not two Néron models. The parent names `Tate_algorithm_at_29`, `Frey_Neron_conductor`, and `Frey_conductor_29_is_Neron` are *theorems in this namespace*: each re-exports the inhabited conjunction `Tate_algorithm_at_2_and_29`. On the parent file `Tate_Frey_Conductor_29.lean` those three names stay `def Prop`.

Honesty note 2.2. The phrase “final $N_E = 928$ ” means the displayed identity $2^5 \cdot 29 = 928$. It is not Tate’s algorithm as a function on integral models. SAGE `sagemath/tate_neron_29.sage` uses the displayed Frey (1), not the short model.

Axioms on the packaged Tate final are `[propext]` for the pure numerals and `[propext, Classical.choice, Quot.sort]` for the conjunction that mentions valuations already inhabited upstream. There is no `Lean.ofReduceBool`, no `sorryAx`, and no new axiom.

2.1. Continued Tate-closure boundary. The companion module `TateCurve.Neron32_928InertiaQ13` at commit `bb4980fb0dba2c2d51c8d928935ffa647d786754` does not reinterpret the preceding numerals as geometry. It records the required local argument in the following forward order:

full $j(q)$ inversion $\longrightarrow E_q/\mathbb{Q}_{29} \longrightarrow j(E_q) = j(E_{A,B}) \longrightarrow I_{26k} \longrightarrow N_{\text{Néron}} \in \{32, 928\} \longrightarrow q^{1/13}$ inertia $\longrightarrow \bar{\rho}_{E,13}|_{I_{29}}$ where $k = v_{29}(B + 3)$ and the curve-attached parameter is required to satisfy

$$v_{29}(q) = v_{29}(\Delta) = 26v_{29}(B + 3).$$

The exact Kodaira equality is a projection from a `TateAlgorithmAt29` witness. The $32/928$ equalities are projections from constructed Néron-conductor data. The cyclotomic formula is a field of the $q^{1/13}$ inertia evidence, not a consequence manufactured from divisibility alone.

The numerical surface

$$13 \mid v_{29}(\Delta), \quad 928/29 = 32, \quad 32 \cdot 29 = 928, \quad 928 \neq 32, \quad 29 \nmid 32$$

is proved independently. The theorem `tate_mod13_unramifiedness_forward` preserves that surface while projecting unramifiedness from the explicit `TateNeron32_928InertiaQ13` boundary. Thus no numerical equality is used backwards to claim Tate uniformization, Tate’s algorithm, a Néron model, a residual representation, or the cyclotomic inertia formula.

3. MAZUR — NO RATIONAL 13-ISOGENY

The file `Mazur_X0_13_No_Isogeny_final.lean` (2f4e16e; incremental 35488699511, 252s) displays the Mazur constants for $X_0(13)$ [11].

Display 3.1 (Mazur numerals).

$$\begin{aligned} |\mathrm{SL}_2(\mathbb{F}_{13})| &= 13 \cdot 12 \cdot 14 = 2184 && (\mathrm{SL}_2_F13_card), \\ 48 &< 2184 && (\mathrm{forty_eight_lt_2184}), \\ 13 &= 2^2 + 3^2 && (\mathrm{thirteen_eq_two_squares}), \\ 288/48 &= 6 && (\mathrm{two_eighty_eight_div_forty_eight}), \\ (\{0, 1\} : \mathrm{Finset} \mathbb{N}).\mathrm{card} &= 2 && (\mathrm{displayed_cusps_card}). \end{aligned}$$

The first four identities are axiom-free `decide` theorems. The cusp-card theorem uses `[propext, Quot.sound]`.

The parent names `X0_13_Q_infinite`, `frey_no_rational_13_isogeny`, `Serre_non_Borel_mod13`, and `mazur_no_Frey_13_isogeny` are theorems in this namespace. Each re-exports the inhabited conjunction of Display 3.1 with the genus `Nat 0`. On the parent file `Mazur_X0_13_No_Isogeny.lean` those four names stay `def Prop`.

Honesty note 3.2 (Genus 0 versus two cusps). $X_0(13)$ has genus 0, so $X_0(13)(\mathbb{Q})$ is infinite over $\mathbb{Q}$. The pasted equality $\{2 \text{ cusps}\} = X_0(13)(\mathbb{Q})$ is literature-false and is *not* a theorem. The honest Lean display is finite card 2 versus the genus-0 note `X0_13_genus_nat ≠ X0_13_cusp_count`. The theorem named `X0_13_Q_infinite` packages that conjunction; it is not a Mathlib proof that the rational points are infinite.

Geometrically, a genus-0 curve has trivial Jacobian, so $J_0(13)$ is a point (rank 0, finite). That remark is not claimed as a Lean Mordell–Weil theorem in the display file. Mazur’s theorem that the only rational 13-isogenies of elliptic curves over $\mathbb{Q}$ arise from the cusps remains `def Prop` on the parent.

Axioms: numerals `[propext]` or axiom-free; package `[propext, Quot.sound]`; LLL no-go persistence `[propext, Classical.choice, Quot.sound]`. No `native_decide`, no `sorryAx`, no new axiom.

3.1. Point-level $X_0(13)$, Frey j , and the Borel boundary. The companion module `MazurCurve.X0_13FreyJBorel` at commit `0a52ba8af887724bcd36df6433a599452e6eee74` records the argument in forward construction order. Its type `FrickeCusp` has exactly two constructors, but this exact cusp table is not identified with all of $X_0(13)(\mathbb{Q})$. A value of `X0_13_Moduli_Data` instead supplies a concrete point-level object, a nonzero Fricke parameter t , and the equality

$$j(t) = \frac{(t^2 + 5t + 13)(t^4 + 7t^3 + 20t^2 + 19t + 1)^3}{t}.$$

Likewise, `FreyJInvariantData` attaches to that construction the exact identity

$$j(E_{A,B}) = \frac{c_4(E_{A,B})^3}{\Delta(E_{A,B})}.$$

The proposition `BorelReducibilityForward` is the required reducibility/Fricke-image equivalence, not a consequence of $48 < 2184$. The theorem `frey_no_rational_13_isogeny_forward` takes this equivalence and the separate Frey-specific exclusion of rational Fricke parameters under $29 \mid B + 3$. Only then does it project the absence of a rational cyclic 13-isogeny. Thus the proved genus, cusp, discriminant, valuation, image-cardinality, and $32/928$ numerals never manufacture the missing geometry or global argument.

4. RIBET — $928 \rightarrow 32$ LEVEL LOWERING

The file `Ribet_Level_Lowering_29_to_32_final.lean` (5631fd1; incremental 35489274297, 282s) displays the residual-level arithmetic of Ribet [12] without inhabiting abstract level lowering.

Display 4.1 (Ribet numerals).

$$\begin{aligned} 928/29 &= 32, & 32 \cdot 29 &= 928, & \neg 29 \mid 32, \\ 2^4 &= 16, & 16 &\mid 32, \\ \dim S_2(\Gamma_0(32))^{\text{disp}} &= 1, & \dim S_2(\Gamma_0(16))^{\text{disp}} &= 0, \\ 32 \cdot \frac{3}{2} &= 48, & 2 \cdot 48/12 &= 8, \\ (\emptyset : \text{Finset } \mathbb{N}).\text{card} &= 0. \end{aligned}$$

The displayed full/new dimensions and the Sturm pair are the LMFDB pin `full=1, new=1, level16=0, sturm=8` for the unique newform 32a1 [9].

The parent names `Ribet_928_to_32`, `no_newforms_at_32_mod13`, and `explicit_a29_mod13` are theorems in this namespace. Each re-exports `no_newforms_at_32`: the numerals of Display 4.1 together with `displayed_frey_matching_newforms.card = 0`. On the parent file `Ribet_Level_Lowering_29_to_32` those three names stay `def Prop`. The theorem named `explicit_a29_mod13` does not compute a_{29} .

Honesty note 4.2 (Newspace dimension 1, not 0). LMFDB records $\dim S_2(\Gamma_0(32))^{\text{new}} = 1$, with unique newform 32a1. The pasted Sage line `CuspForms(32,2).new_subspace().dimension()` is `0` is *not* a theorem. The honest Lean display is the empty matching Finset (card 0) versus displayed dimension 1. Witnesses: `sagemath/ribet_no_newforms_32.sage` and `certs/ribet_no_newforms_32.json` pin `full=1, new=1, level16=0, sturm=8`.

Axioms: `display [propext, Quot.sound]`; LLL no-go `[propext, Classical.choice, Quot.sound]`. No `Lean.ofReduceBool`, no `sorryAx`, no new axiom.

Forward Ribet closure. The companion module `RibetCurve.Ribet32a1KrausA29` at commit `492a29293c1fcd72b4` packages the geometric and automorphic inputs explicitly. Its forward theorems project Wiles/BCDT modularity, concrete residual irreducibility, unramifiedness at 29, Kodaira type I_{26k} , local conductor exponent one, the global Néron 32/928 split, a residual level-32 eigenform, and exact identification with 32a1. At good reduction it records the mismatch with $a_{29}(32a1) = -10$; complete bad-reduction residue elimination remains a separate explicit field.

The proved surface

$$\Delta = 16A^8B^8(A^4 + B^4)^2, \quad v_{29}(\Delta) = 26v_{29}(B + 3), \quad 13 \mid v_{29}(\Delta),$$

together with $928/29 = 32$, $32 \cdot 29 = 928$, $|\text{SL}_2(\mathbb{F}_{13})| = 2184$, and $48 < 2184$, is never used backwards to construct modularity, a representation, irreducibility, conductor data, an eigenform, or elimination. The new theorem dependencies are subsets of `[propext, Classical.choice, Quot.sound]`.

5. KOLYVAGIN — MW RANK-0 DISPLAY ON 26a1, 26b1

The file `Kolyvagin_MW_Rank0_26a1_26b1_final.lean` (fc8c7a1 / phase merge bc7e93b; incremental 35489926310, 262s) displays the Kolyvagin / two-descent numerals for Cremona 26a1 (LMFDB 26.a2) and 26b1 (LMFDB 26.b2) [8, 1, 9].

Display 5.1 (Kolyvagin numerals and typeclass upgrade).

$$\begin{array}{ll} |\text{Sel}_2|^{\text{disp}} = 1 & (\text{Sel2_card_one}, \text{displayed_Sel2_card_26a1/b1}), \\ 3 \cdot 7 = 21 & (\text{three_mul_seven}), \\ L/\Omega = 1/3 \text{ on } 26a1, \quad 1/7 \text{ on } 26b1 & (\text{norm_num on } \mathbb{Q}), \\ \neg \text{IsRankZero on both models}, & \\ \text{Nonempty}(\text{Fintype } \alpha) \rightarrow \text{Finite } \alpha & (\text{finite_iff_nonempty_fintype}), \\ |\text{Unit}| = 1 & (\text{Subsingleton card on Unit}). \end{array}$$

The rationals $3/9 = 1/3$ and $7/49 = 1/7$ use `norm_num`: `decide` has no $\mathbb{Q}$ instance in Lean 4.12.

Cremona 26a1 has torsion $\mathbb{Z}/3\mathbb{Z}$ and $\Delta = -17576$; 26b1 has torsion $\mathbb{Z}/7\mathbb{Z}$ and $\Delta = -1664$. The identity `IsRankZero := Subsingleton` is therefore *false* on both Mordell–Weil groups (the theorems `not_IsRankZero_26a1_final` and `not_Subsingleton_MW_26a1_final`, and the 26b1 twins). The `Subsingleton → Fintype` path applies to `Unit`, not to $E(\mathbb{Q})$.

The parent names `MW_rank_zero_fintype`, `Kolyvagin_L_nonzero_imp_MW_rank_zero`, and `TwoDescent_implies` are theorems in this namespace. Module-named theorems `TwoDescent_26a1_26` (from 7c19ad0),

BSD_MordellWeil, J0_26_BSD_26a1_26b1, and X0_26_Full2Torsion package the same numerals. On the parent files, MW_rank_zero_fintype, Kato, BSD_MordellWeil.MW_rank_zero, and X0_26_Full2Torsion.J0_26_rank0 stay `def Prop`.

Honesty note 5.2 (Not $E(\mathbb{Q})$ finite). Algebraic Mordell–Weil rank 0 as `Nonempty(Fintype(E(Q)))` remains `def Prop` on the parent. The packaged theorem `MW_rank_zero_fintype` is Display 5.1, not Kato–Kolyvagin and not Sage `E.rank()`. Witness: `sagemath/kolyvagin_rank0_26.sage` with `certs/kolyvagin_rank0_26.json`.

Axioms: `[propext, Classical.choice, Quot.sound]`. No `Lean.ofReduceBool`, no `sorryAx`, no new axiom.

Forward Kolyvagin closure. The companion module `KolyvaginCurve.KolyvaginL_OmegaSel2Fintype` at commit `e0d538ac2f344a75586fe2745e5867944ec8e429` begins with curve-attached L -functions, positive periods, and exact normalized-value identifications. Only then does it derive analytic nonvanishing and apply an explicit Kato–Kolyvagin boundary. Independently, it begins with constructed two-Selmer exact sequences, exact cardinality identifications, and Mordell–Weil finite generation before applying the singleton-Selmer boundary.

Thus $L/\Omega = 1/3$ for 26a1, $L/\Omega = 1/7$ for 26b1, $|\text{Sel}_2| = 1$, and $2^{\dim} = 1$ are forward projections, not constructors. Actual `Nonempty (Fintype E(Q))` witnesses are explicit data and coexist with the proved `¬Singleton E(Q)` facts from nonzero torsion. The new theorem dependencies are subsets of `[propext, Classical.choice, Quot.sound]`.

6. LEAN FORMALIZATION — EXPLICIT TYPES

Every theorem in the four finals carries an explicit type. Lean 4.12 rejects `theorem foo := bar`. The required opens are `open Nat Finset Classical`. Typical types are $\mathbb{N}$, `Prop`, `Finset ℕ`, and `Fintype`. Proofs use `decide` (or `norm_num` on $\mathbb{Q}$), never `native_decide`, so `Lean.ofReduceBool` does not appear.

Pin. Toolchain `leanprover/lean4:v4.12.0`. Mathlib `809c3fb3b5c8f5d7dace56e200b426187516535a`, `inputRev v4.12.0`. Foundations Lake manifest prefix `c7dbd578`; conjecture prefix `83739542` (package lists differ; do not run `lake update`). Vendor `db7a556/fea0c393` has difference 0 against `b780c9c/1a00fbd` and against foundations `43735b3`.

Axioms. Allowed axiom lists are `[propext]`, `[propext, Quot.sound]`, and `[propext, Classical.choice, Quot.sound]`. There is no `sorryAx` and no new axiom. Parent files keep the algebraic names as `def Prop`. Display files contain no new `def Prop`.

Integrated conditional bridge. The companion release registers the full Tate, Mazur, Ribet, and Kolyvagin surfaces together with `Conditional32Newform.Bridge` and `RequiresTateMazurRibetFor32`. Doc. The bridge proves

$$\Delta = 16A^8B^8(A^4 + B^4)^2, \quad v_{29}(\Delta) = 26v_{29}(B + 3), \quad 13 \mid v_{29}(\Delta),$$

as well as $928/29 = 32$ and $32 \cdot 29 = 928$. Its theorem `has_newform_at_level_32_of_assumptions` takes exactly `hMazur`, `hTate`, and `hRibet`. The result is `HasNewformAtLevel32`, a proposition-valued evidence record. Wiles/BCDT modularity, the geometric mod-13 representation and absolute irreducibility, Tate/Néron and cyclotomic-inertia theory, genuine Ribet lowering, identification with 32a1, and complete Kraus elimination remain explicit.

Forward Conditional-32 closure. The module `Conditional32Newform.BridgeForward` at commit `75d546a56de5ff901faf50611c5fd3d449612aab` first projects the concrete Mazur conclusion from point-level $X_0(13)$, Borel/Fricke, and global-exclusion evidence; the concrete Tate conclusion from uniformization, Néron, $q^{1/13}$ inertia, and mod-13 unramifiedness; and the concrete Ribet conclusion from residual 32a1, $a_{29} = -10$, and complete Kraus elimination.

The newer concrete modules and the older Conditional-32 interface use distinct proposition names. Three explicit typed conversion boundaries are therefore fields of `Conditional32BridgeData`;

they are neither silently identified nor inferred from numerical equalities. Only after those conversions does `three_hypotheses_forward` produce the Mazur, Tate, and Ribet hypotheses. Exactly those three hypotheses feed `has_newform_at_level_32_of_assumptions_forward` and its `HasNewformAtLevel32Data` result.

Integrated forward dependency table. The module `RequiresTateMazurRibetFor32.ForwardDoc` at commit `70bdf8149b377861a1504ef7d4932711e15e6905` closes the documentation surface without erasing its hypotheses. Its record orders `TateCurve.ClosureContinued`, `TateCurve.Neron32_928InertiaQ13`, `MazurCurve.X0_13FreyJBorel`, `RibetCurve.Ribet32a1KrausA29`, `KolyvaginCurve.KolyvaginL_OmegaSel2Fintype`, and `Conditional32Newform.BridgeForward`. The exact-dependency theorem returns all six boundaries; the integrated theorem returns that dependency table together with the conditional level-32 evidence.

The Kolyvagin interface remains logically separate from the three bridge hypotheses, even though it is preserved in the release ledger. The final bridge still consumes only Mazur, Tate, and Ribet, and it does so only after the three explicit typed conversions. The parallel numerical theorem records Δ , $v_{29}(\Delta)$, $928/29$, $|\mathrm{SL}_2(\mathbb{F}_{13})|$, a_{29} , L/Ω , and Selmer cardinalities without using any of them backwards to construct the six interfaces.

Final Beal interface. The module `BealFinal.Main` at commit `995b9c4810869c1ce6531b32080e668c1f5ff233` extends the exact dependency ledger without changing its formal status. The record `BealFinalData` carries the six-stage `Requires` package, the equalities `C1_floor=143186215390` and `B0_nat=1000000`, the full $B \leq 10^6$ search proposition, and a separate `BealTheoremData` field. The latter states that every positive solution with all exponents at least three has a prime dividing all three bases.

The projection theorems expose that common-prime conclusion but do not prove it from the signature $(4, 4, 13)$ numerics. Likewise, the imported finite search proves only $B < 1000$; the full $B \leq 10^6$ statement remains an explicit field. The final numerical surface records the established discriminant, valuation, level, newform, normalized-value, and Selmer equalities together with the two constants and the proved $B < 1000$ slice.

`v25` forward retrofit. The post-final module `BealMatveevBealV25B0Search.Forward` orders three interfaces without strengthening them. `MatveevBoundData` carries `C1_floor=143186215390` and `B0_nat=1000000`. `Level26SearchData` carries only the forward numerical surface $\Delta = 16A^8B^8(A^4 + B^4)^2$, $v_{29}(\Delta) = 26v_{29}(B+3)$, $13 \mid v_{29}(\Delta)$, $928/29 = 32$, and $32 \cdot 29 = 928$. `MatveevLevel26ForwardData` then carries the proved $B < 1000$ elimination and, separately, the full $B \leq 10^6$ proposition.

The three projection theorems consume those fields in that order. They do not infer a Matveev bound, a level-lowering construction, or the full search from the numerical identities. Their dependency lists are exactly `[propext, Classical.choice, Quot.sound]`; the historical finite-computation dependency of the existing small-search theorem is not introduced into the new projections.

Forward Tate closure. The subsequent module `TateCurve.Neron32_928InertiaQ13` projects the exact symbol I_{26k} from Tate-algorithm evidence, the $32/928$ split from constructed Néron data, and $v_{29}(q) = v_{29}(\Delta)$ from curve-attached uniformization data. Its final mod-13 theorem consumes the proved numerical surface but obtains unramifiedness only from the explicit Tate–Néron– $q^{1/13}$ inertia boundary. In particular, $13 \mid v_{29}(\Delta)$ is never used backwards to construct the geometric representation or cyclotomic formula.

Forward Mazur closure. The subsequent module `MazurCurve.X0_13FreyJBorel` keeps the two-constructor Fricke cusp table distinct from the infinitely many rational points on the genus-zero parameter line. It projects the exact Fricke function from point-level moduli data and $j = c_4^3/\Delta$ from curve-attached Frey data. Its no-rational-13-isogeny theorem consumes both the explicit Borel/Fricke equivalence and the Frey-specific global exclusion. Neither genus and cusp numerals nor image-cardinality numerals are used backwards to construct moduli, geometric torsion, a Galois representation, or Borel containment.

Forward Ribet closure. The subsequent module `RibetCurve.Ribet32a1KrausA29` takes constructed modularity, residual-representation, irreducibility, inertia, conductor, Néron, residual-eigenform,

and 32a1 data. It projects those data in the mathematical direction and keeps the good-reduction $a_{29} = -10$ mismatch distinct from complete bad-reduction Kraus elimination. The independent numerical surface includes the discriminant and valuation identities, $928/29 = 32$, $32 \cdot 29 = 928$, $|\mathrm{SL}_2(\mathbb{F}_{13})| = 2184$, and $48 < 2184$; none of those numerals inhabits a geometric or automorphic boundary.

Forward Kolyvagin closure. The subsequent module `KolyvaginCurve.KolyvaginL_OmegaSel2Fintype` keeps the analytic and two-descent routes explicit. Constructed L -functions, positive periods, and normalized-value identifications precede Kato–Kolyvagin. Constructed Selmer exact sequences, exact cardinality identifications, and Mordell–Weil finite generation precede the singleton–Selmer implication. The displayed values $1/3$, $1/7$, $|\mathrm{Sel}_2| = 1$, and $2^{\dim} = 1$ never construct those inputs. Actual `Fintype` witnesses provide the finite-point endpoint while the nonzero torsion points continue to refute the older `Subsingleton` encoding.

CI, incremental only. Foundations `build.yml` is `HonestB0Search incremental: lake exe cache get, then +Tate_Frey_Conductor_29_Neron_final, +Mazur_X0_13_No_Isogeny_final, +Ribet_Level_Lowering, +Kolyvagin_MW_Rank0_26a1_26b1_final` (~ 40 s on a warm `.lake`), then `lake build HonestB0Search` and `lake build Level126` (1–2s each). Cache key `lake-${hashFiles(lean-toolchain,lake-manifest.json)}`–`Representative greens: 35488239286` (137s, 11c0ad7); 35488699511 (252s); 35489274297 (282s); 35489926310 (262s); phase 35490246507 (303s) and 35524223408; About chore 35552037733 (01:47:20Z–01:52:05Z, 285s, `.lake` restored). On those working branches `main.yml` recorded 0 runs. The phase incremental 35524223408 is the green gate. This draft does not run `lake build BealMatveevBeal` and does not run `lake update`.

The integrated release ran the larger reproducibility gate: cache restoration of 5,134 files and a combined build of 5,206 jobs, including `BealMatveevBealV25B0Search`, all four `BlockerDoc` roots, the conditional bridge, and its requirements document. The publication-surface validator and `verify-matveev-beal.sh` passed. Its axiom report is `[propext, Classical.choice, Quot.sound]` with no `sorryAx`.

Conjecture `main.yml`. On `beal-conjecture` the hours job `build+HonestB0Search+Level126` runs only on `main` (and matching merge/phase/v24–v27 branches). Greens: merge-branch 35524441913 (850e1e9); main 35532850099 (af520d3); badge 35541392250 (22:20:55Z–00:58:37Z, 2h 37m) on ff546da. Tip ea261b9 is the About chore on top of that badge.

7. COMPUTATIONAL VERIFICATION

Zenodo. The integrated software version DOI 10.5281/zenodo.22922473 records the v25.0.1 post-final forward retrofit [3]. The preceding software version is 10.5281/zenodo.22912430 [4]. The concept DOI 10.5281/zenodo.22379293 stays, recorded as `isVersion0f` [2]. The corresponding foundations GitHub release is at 385ff85 [5]; the predecessor monorepo tag v30.0.0-level-26-structural-verification remains at af520d3.

Sage and JSON.. The Lean displays are paired with static Sage scripts and JSON pins, not with a Sage kernel in CI: `sagemath/tate_nero_29.sage`; `sagemath/ribet_no_newforms_32.sage` with `certs/ribet_no_newforms_32.json` (`full=1, new=1, level16=0, sturm=8`); `sagemath/kolyvagin_rank0_26.sage` with `certs/kolyvagin_rank0_26.json`. LMFDB labels used as pins are 32a1, 26a1, and 26b1 [9]. Repository surface. Folder READMEs are complete outside the vendor tree (difference 0 against 43735b3). GitHub About texts are 302 characters (foundations, homepage the concept DOI) and 308 characters (conjecture, homepage the version DOI). Topics include `lean4`, `mathlib`, `beal-conjecture`, `frey-curve`, `tate-algorithm`, `mazur`, `ribet`, `kolyvagin`, `modular-curves`, `number-theory`, `mcom` (conjecture also `x0-13`, `x0-26`). There is no arXiv badge.

Tags and working branches. Foundations tags v0.27, v0.28, v0.29, v0.30 stay live. Phase `phase-darmon-merel-4` stays 2b60ccd (v0.30-mcom-structural-verification). Working finals stay `tate-neron-final-v29` 11c0ad7, `mazur-no-isogeny-final-v29` 2f4e16e, `ribet-928-to-32-final-v29` 5631fd1, `kolyvagin-rank0-final-v29` 11c0ad7.

fc8c7a1. Conjecture tag v30.0.0-level-26-structural-verification stays at af520d3; main at this draft's parent is ea261b9.

8. CONCLUSION

Level 26 structural verification records four honest formal surfaces: Tate $928 = 2^5 \cdot 29$ (I_0^* , I_8), Mazur 2184 and cusp card 2 versus genus 0, Ribet $928/29 = 32$ with matching Finset $\emptyset$ against LMFDB 32a1 of dimension 1, and Kolyvagin $|\text{Sel}_2| = 1$, $3 \cdot 7 = 21$, $L/\Omega = 1/3$ and $1/7$ on 26a1/26b1. The software versions are v30.0.0-level-26-structural-verification on beal-conjecture (af520d3, About tip ea261b9) and v30.1.7-beal-final-43735b3 on foundations 995b9c4, followed by the post-final retrofit v25.0.1-forward-43735b3 at 385ff85. The preceding software DOI is 10.5281/zenodo.22912430; the monorepo predecessor remains v30.0.0-level-26-structural-verification Phase stays 2b60ccd. Tags v0.27–v0.30 stay live. The bridge is conditional on the named Mazur, Tate, and Ribet inputs; the analytic Kolyvagin line remains separate. Parent algebraic statements remain explicit proposition boundaries. The general common-prime theorem and the full $B \leq 10^6$ search are explicit fields of the final interface, not consequences extracted from its numerical surface. The retrofit likewise projects the Matveev constants, level-26 arithmetic, and proved $B < 1000$ slice forward while retaining $B \leq 10^6$ as an explicit field.

This draft is a Mathematics of Computation submission, not an arXiv preprint. Cite the version DOI 10.5281/zenodo.22922473 and the concept DOI 10.5281/zenodo.22379293.

REFERENCES

1. John E. Cremona, *Algorithms for modular elliptic curves*, 2 ed., Cambridge University Press, 1997.
2. David Fox, *Beal level 26 foundations (concept)*, Zenodo, 2025, Concept DOI; isVersion0f for the v30 record.
3. ———, *Formal verification of the level 26 foundations for signature (4,4,13)*, Zenodo, 2026, Software version v25.0.1-forward-43735b3; post-final forward Matveev and level-26 retrofit retaining the full $B \leq 10^6$ search as an explicit boundary.
4. ———, *Formal verification of the level 26 foundations for signature (4,4,13)*, Zenodo, 2026, Software version v30.1.7-beal-final-43735b3; final forward interface with the full search and general common-prime theorem retained as explicit proof boundaries.
5. ———, *beal-level-26-foundations tag v25.0.1-forward-43735b3*, <https://github.com/DavidFox998/beal-level-26-foundations/releases/tag/v25.0.1-forward-43735b3>, 2026, Commit 385ff85c78c351de83b3fd625a208c9dea003647; exact merge base 43735b3b14a9e844d4b4bdf57fa5f68d29e749ee.
6. ———, *beal-level-26-foundations tag v30.1.7-beal-final-43735b3*, <https://github.com/DavidFox998/beal-level-26-foundations/releases/tag/v30.1.7-beal-final-43735b3>, 2026, Commit 995b9c4810869c1ce6531b32080e668c1f5ff233; exact merge base 43735b3b14a9e844d4b4bdf57fa5f68d29e749ee.
7. Gerhard Frey, *Links between stable elliptic curves and certain Diophantine equations*, Annales Universitatis Saraviensis. Series Mathematicae **1** (1986), no. 1, 1–40.
8. Victor A. Kolyvagin, *Euler systems*, The Grothendieck Festschrift, Vol. II **87** (1990), 435–483.
9. LMFDB Collaboration, *The L-functions and modular forms database*, <https://www.lmfdb.org>, 2026, Labels 32a1, 26a1 (LMFDB 26.a2), 26b1 (LMFDB 26.b2).
10. R. Daniel Mauldin, *A generalization of Fermat's last theorem: The Beal conjecture and prize problem*, Notices of the American Mathematical Society **44** (1997), no. 11, 1436–1437.
11. Barry Mazur, *Rational isogenies of prime degree*, Inventiones Mathematicae **44** (1978), no. 2, 129–162.
12. Kenneth A. Ribet, *On modular representations of $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ arising from modular forms*, Inventiones Mathematicae **100** (1990), no. 2, 431–476.
13. Jean-Pierre Serre, *Sur les représentations modulaires de degré 2 de $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$* , Duke Mathematical Journal **54** (1987), no. 1, 179–230.
14. John Tate, *Algorithm for determining the type of a singular fibre in an elliptic pencil*, Modular Functions of One Variable IV, Lecture Notes in Mathematics, vol. 476, Springer, Berlin, 1975, pp. 33–52.